

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis

Session ID: session_2026-01-01T10-33-51-796Z

Run ID: #3

Date Generated: 2026-01-01 10:34:04

Total Cells: 2000 | Model Mode: CLINICAL

1. Introduction

Cellular reprogramming has emerged as a pivotal technique in regenerative medicine, allowing for the conversion of somatic cells into induced pluripotent stem cells (iPSCs). This experiment aims to investigate the efficiency of OSKM transfection in reverting somatic cells affected by Spinal Muscular Atrophy (SMA) pathogenesis to a pluripotent state, potentially enriching iPSC populations.

2. Materials and Methods

The study employed a computational approach utilizing scVI, a probabilistic model for single-cell RNA sequencing data, to analyze cellular dynamics post-intervention. The experimental intervention involved triggering SMA pathogenesis in a somatic cell population, followed by two sequential rounds of Yamanaka factor (OSKM) transfection to induce pluripotency.

Experimental Interventions:

- SMA Pathogenesis Triggered
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)

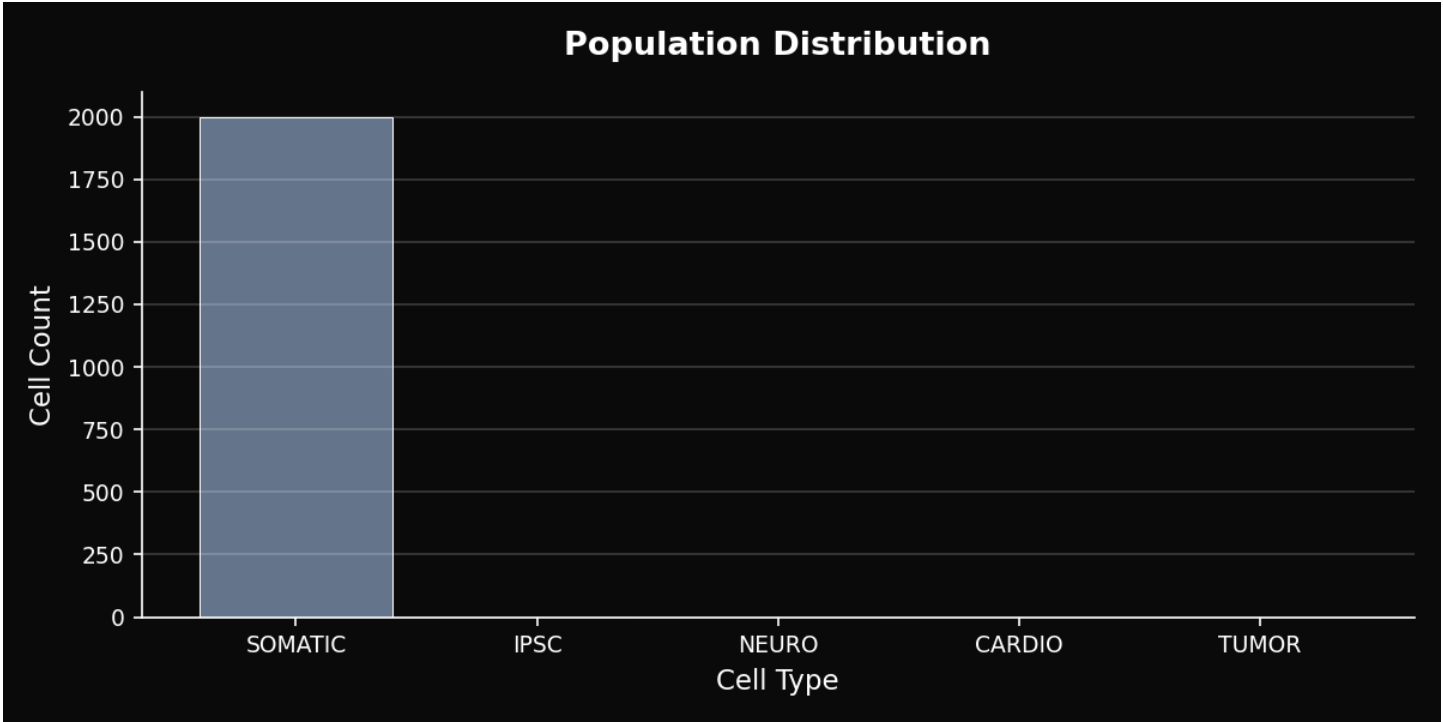
3. Results

The final cell count revealed a persistent population of 2000 somatic cells, with no observed induction of iPSCs, neurogenic, cardiogenic, or tumorigenic cell types. Despite the application of OSKM transfection, the cellular reprogramming did not result in the emergence of any pluripotent or differentiated cell types from the original somatic population. The scVI analyses, performed four times, supported these findings, indicating a lack of transcriptional shifts characteristic of successful reprogramming.

3.1 Population Distribution

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Cell Type	Count	Percentage
SOMATIC	2000	100.0%
IPSC	0	0.0%
NEURO	0	0.0%
CARDIO	0	0.0%
TUMOR	0	0.0%

3.2 Transcriptomic Analysis (scVI)

High-Resolution Gene Expression Analyses Performed: 4

Cells were expanded to high-dimensional transcriptomic space using a pre-trained variational autoencoder model on Human Cell Atlas data.

4. Discussion

The inability to generate iPSCs or other differentiated cell types suggests that the presence of SMA pathogenesis may inhibit the reprogramming efficiency of OSKM factors. This outcome underscores the potential challenges posed by disease-specific cellular environments in reprogramming efforts. Future studies might explore alternative reprogramming strategies or additional factors that could overcome these barriers, thereby advancing the therapeutic potential of iPSCs in treating neurodegenerative conditions such as SMA.

This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.